D15 Cellular Drift Eliminator

D15 CELLULAR DRIFT ELIMINATOR PRODUCT SPECIFICATION

The drift eliminators are to be manufactured by Brentwood Industries and will meet the following specifications:

1. Scope:
   ACCU-PAC® D15 cellular-type drift eliminator is designed to remove entrained water droplets from exit air streams of cooling towers at minimum pressure loss in either counter-flow or cross-flow configurations.
2. Material
   a. General
      The drift modules shall be fabricated from rigid, corrugated PVC sheets that are conducive to cooling water and UV protected. The drift modules shall be resistant to rot, fungi, bacteria and inorganic/organic acids and alkalies as commonly found in cooling towers.
   b. PVC Raw Material
      i. The PVC raw material shall be prime, rigid PVC conforming to commercial standard ASTM D1784:12344B to 12454B and meet the Cooling Technology Institute's plastic material standard, CTI STD-136.
      ii. The before forming PVC sheets shall be of uniform thickness and free from holes, air bubbles, foreign matter, undispersed raw material or other manufacturing defects which may adversely affect their performance.
3. Panel Construction
   a. The eliminators shall be of the cellular, multi-pass type with 30mil (0.76mm) flat sheets uniformly spaced within the panel to provide structural stiffness and durability. The sheets shall be assembled by use of solvent cements which shall not be adversely affected by the wet cooling tower environment. The solvent cement will be applied in such a way as to not create hardened glue residue within the panel's air flow passageways. The panels shall be constructed so that the individual panels can nest together, forming one integrated drift eliminator section and maximizing drift eliminator installation effectiveness. Panels with flat side stiffener sheets may be provided only if fully caulked.
   b. The recommended drift module span is related to the number of stiffener sheets assembled within the panel. The support span must not exceed that specified by the manufacturer for the number of stiffener sheets. For Brentwood's D15 support recommendations, see Brentwood drawings CTP-0176 through CTP-0179.
   c. Material used for the eliminators shall be suitable for continuous operation at the maximum specified discharge air temperature
   d. The drift modules shall measure up to 24 inches wide, 5 1/4" in depth and up to 10 ft. long.
4. Performance
   a. The drift eliminators shall be laboratory tested and have a drift rate of no more than 0.003% of the circulating water-flow rate as determined by the CTI STD-140 HBIK drift test method.
   b. A certified drift test report of testing conducted by an independent lab, shall be provided that shows that the drift eliminator will meet the specified level of performance.
   c. Eliminators shall be fire resistant with a maximum flame spread index (FSI) of 15 or less according to ASTM E-84. A certified test report showing that the cellular eliminator will meet this flame spread rate shall be provided.
   d. Pressure drop curves shall be provided showing the operating (wet) pressure drop as a function of air velocity.
5. Installation
   a. The drift eliminators shall be installed in accordance with Brentwood's installation guidelines to ensure maximum drift reduction.